JEPA-x: Cross-Predictive Physics Grounding for Forecastable Latent Dynamics

Human-centered Robotic Lab

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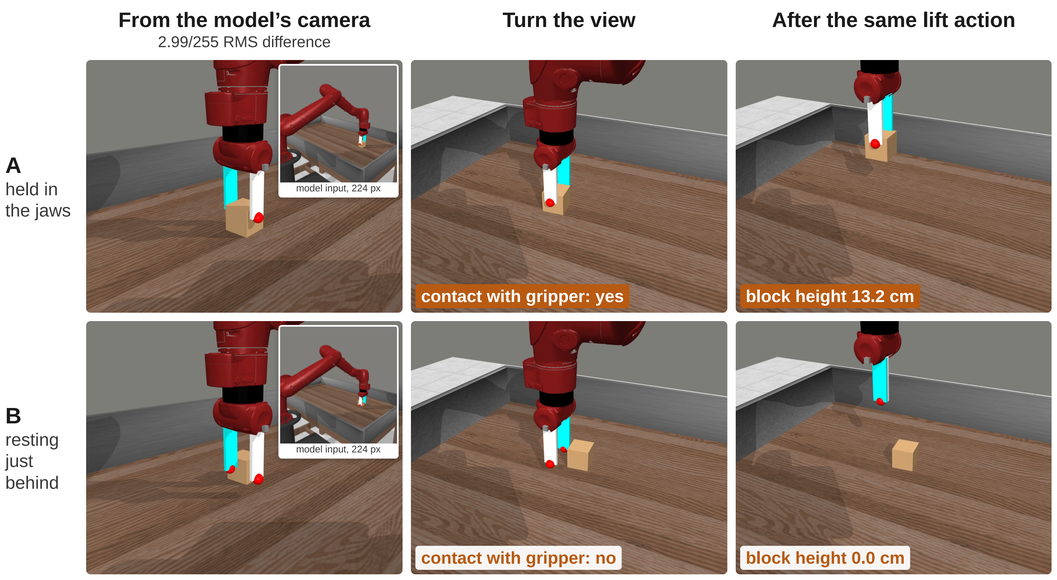
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CoRL

Video, demos & paper
wkh923.github.io/jepa-x
arXiv 2608.24044

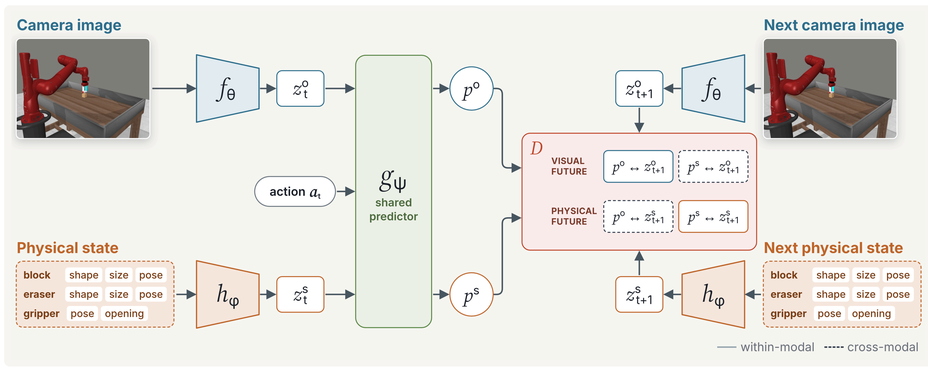
Motivation — Similar pixels, different physics

A visual world model must predict how the scene evolves from pixels alone. Two scenes can look almost identical to the model, yet behave differently under the same action.

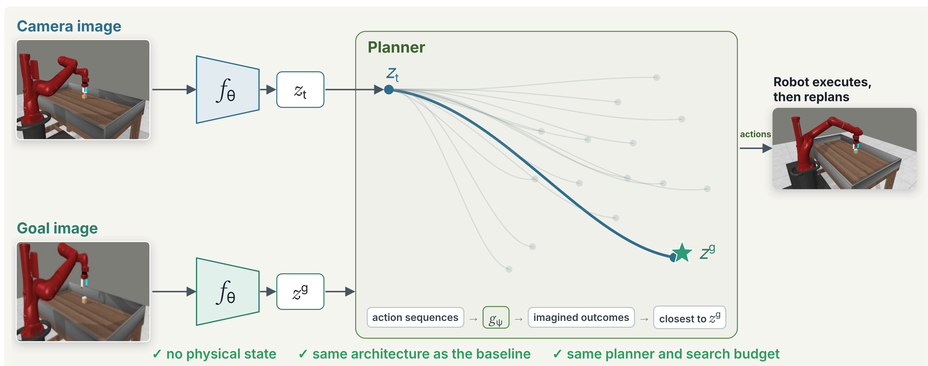


Before the lift, the pixels barely differ, but the physical state already does. So we use physical trajectories to guide training.

Method — JEPA-x: predict both futures from either view



At deployment: plan from pixels



During training, each trajectory comes in two views: camera images and the physical state of the objects and gripper. One shared predictor rolls either view forward under the same action, and each prediction must match **both** futures.

$$\mathcal{L}_{\text{JEPA-}x} = \sum_{m,n \in \{o,s\}} \left\| g_\psi(Z_t^m, a_t) - z_{t+1}^n \right\|_2^2 + \beta \left[\Omega(z^o) + \Omega(z^s) \right]$$

- Within-modal terms keep self-prediction; **cross-modal terms tie the visual latent to how the physical world evolves.**
- Physical state = object tokens (SDF geometry, extents, pose) + gripper: one encoder for all 22 configurations.

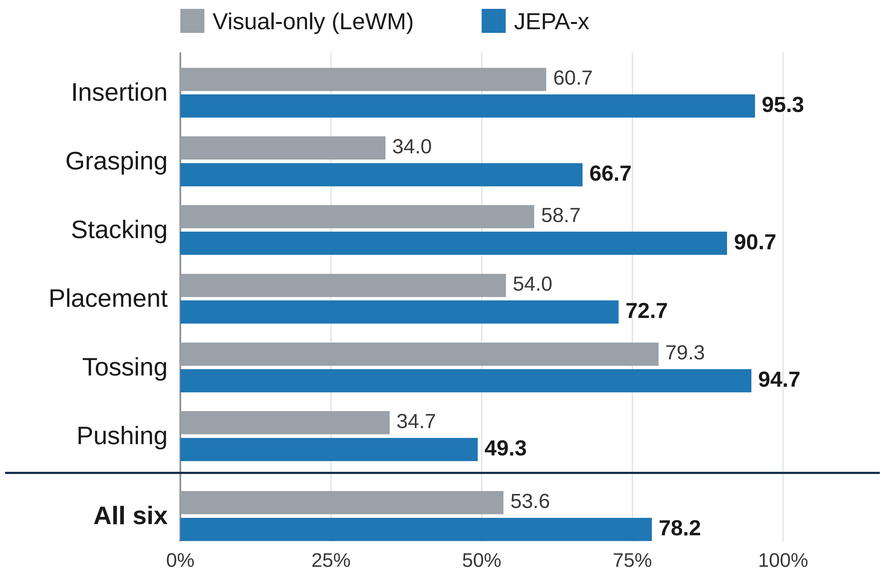
At deployment: the physical branch is removed. The planner samples action sequences, imagines their outcomes with g_ψ, executes the one that ends closest to the goal latent z^g, and replans.

Visual-only baseline: LeWorldModel (LeWM; Maes et al., 2026). JEPA-x keeps its encoder, predictor and planner and adds the training-only physical branch.

Results — Better control from pixels

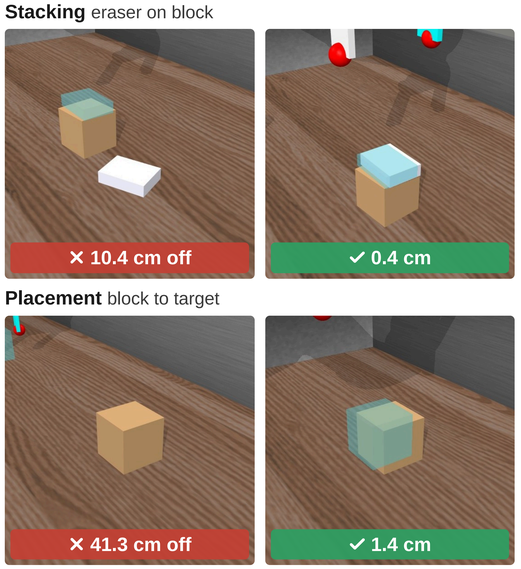
53.6 → 78.2%

mean control success · 6 families · 22 configurations



Same planner and search budget for both; mean of three seeds.

Same start, same goal, same planner



Final close-ups; cyan = goal pose, number = object's distance to it.

Readable is not the same as forecastable

Decodability	Can physical variables be read out of one latent?
Forecastability	Does the latent evolve predictably under action, even for a new predictor?
Control utility	Do predicted outcomes rank candidate actions correctly?

Readable physics is not predictable physics

Objective	Position R^2 ↑ readout	Drift ↓ new predictor	Control ↑ success %
Visual self-prediction	0.978	0.361	53.6
Regress state regression	0.991	0.373	53.7
Distill latent matching	0.991	0.516	57.7
JEPA-x cross-prediction	**0.978**	**0.104**	**78.2**

Regression makes position most readable, yet dynamics are no more forecastable and control stays near the baseline. With shuffled visual–physical pairs, control collapses to 3.6%.

Physical grounding during training, pixels at deployment.

- Drift of a new predictor 0.361→0.104; no extra test-time cost